

ARTICLE

Probabilistic Record Linkage Using Pretrained Text Embeddings

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(Received 19 July 2024; revised 14 March 2025; accepted 8 April 2025; published online 28 August 2025)

Abstract

Pretrained text embeddings are a fast and scalable method for determining whether two texts have similar meaning, capturing not only lexical similarity, but semantic similarity as well. In this article, I show how to incorporate these measures into a probabilistic record linkage procedure that yields considerable improvements in both precision and recall over existing methods. The procedure even allows researchers to link datasets across different languages. I validate the approach with a series of political science applications, and provide open-source statistical software for researchers to efficiently implement the proposed method.

Keywords: probabilistic record linkage; fuzzy string matching; embeddings; large language models (LLMs); GPT-3; GPT-4; active learning; text-as-data

Edited by: Daniel J. Hopkins and Brandon M. Stewart

1. Introduction

Empirical social scientists frequently need to merge information from multiple datasets prior to conducting their analyses, but it is only in rare cases that two datasets contain a shared variable that unambiguously identifies which records belong to the same entity. In the absence of such exact matching variables, researchers must perform *fuzzy record linkage*—linking records based on some measure of similarity between variables. When working with text data, existing approaches commonly rely on *lexical* measures of string similarity (Jaro 1989). These include “edit distance” measures (e.g., Jaro-Winkler and Levenshtein distance), string metrics that compare the frequency distributions of characters (e.g., cosine similarity), and set theoretic measures (e.g., Jaccard similarity), among many others. The most commonly used and cited fuzzy record linkage procedures in political science employ one or more of these metrics to capture the distance between pairs of records (Enamorado, Fifield, and Imai 2019; Kaufman and Klevis 2022).

Lexical similarity is a powerful tool for record linkage when datasets contain misspellings, typos, or other irregularities in data entry. But these measures have well-understood shortcomings, particularly in cases where lexically dissimilar strings can be used to represent the same entity. For example, the name “Patricia” is more lexically similar to “Patrick” than it is to its nickname “Trish.” Many record linkage problems that political scientists encounter have this property, in which semantically similar records can be represented by lexically dissimilar strings. Elected officials may be referenced by their legal name in one dataset and their nickname in another. An organization may be listed under its full name in one dataset and an acronym in another. For scholars of comparative and international politics, records may even appear in multiple languages. When faced with record linkage problems like these, a measure that captures not only the lexical similarity between strings, but their *semantic* similarity as well, would be highly desirable.

Table 1. Examples where lexical similarity is a misleading measure of match quality.

String 1	String 2	Levenshtein	Jaro-Winkler	Jaccard	Embedding
AARP	American Association of Retired Persons	0.103	0.517	0.188	0.837
AARP	AAA	0.500	0.722	0.333	0.555
USPS	U.S. Post Office	0.214	0.655	0.250	0.814
USPS	UPS	0.750	0.806	1.000	0.753
Mike Kelly	George Joseph “Mike” Kelly, Jr.	0.323	0.354	0.421	0.827
Mike Kelly	Mark Edward Kelly	0.471	0.757	0.538	0.615
Kit Bond	Christopher Samuel Bond	0.304	0.475	0.368	0.605
Kit Bond	Katie Britt	0.364	0.627	0.455	0.445

Note: The first row of each pair is the true match, and the best match according to four string similarity measures is in bold. In each case, lexical measures select the wrong match, while the cosine similarity between pretrained text embeddings selects the correct match.

Fortunately, such measures have recently become widely available, thanks to rapid advances in large language models (LLMs) based on the transformer architecture (Vaswani *et al.* 2017). These models encode language using *text embeddings*, wherein each word is represented by a real-valued vector of numbers (Rodriguez and Spirling 2022). Once trained, the distance between these text embeddings provides a useful measure of semantic similarity: words that are closer together in embedding space tend to have similar meaning. Formally, if two strings of text are represented by the vectors **a** and **b**, then their cosine similarity $\frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|}$ measures how semantically related they are—with 0 being completely orthogonal and 1 being identical.

Table 1 provides several examples in which the cosine similarity between text embeddings provides a better measure of match quality than lexical similarity (see the next section for details on how these cosine similarities are computed). Consider, for example, the problem of linking an organization’s full name with its acronym (first four rows). Lexical measures of string distance will struggle with this sort of record linkage task, since an organization’s acronym may be lexically more similar to the acronym of another organization than it is to its own full name! In contrast, embedding vectors can encode the fact that AARP stands for American Association of Retired Persons, by representing those strings as vectors close to one another in space—this is how language models based on such embeddings (e.g., ChatGPT) “know” the relationship between those two concepts. In each of the examples in Table 1, the cosine similarity between text embeddings chooses the correct match, while lexical measures of string similarity do not. Consequently, a record linkage procedure that incorporates this measure of similarity may significantly outperform procedures that rely exclusively on lexical similarity.

This is not the first article to propose using text embeddings for record linkage. Indeed, there is by now an extensive literature applying transformer models to what computer scientists call *entity resolution*—determining whether two or more entries in a large dataset refer to the same entity (Tang *et al.* 2022; Zhou *et al.* 2021). These models have had significant practical applications in areas like e-commerce, where merging product records across multiple websites is a challenging large-scale problem. These approaches have been adapted to social science applications as well, most notably in the work of Arora and Dell (2023). What distinguishes the current article from previous work is that it incorporates embedding similarity into a probabilistic record linkage procedure. Such procedures are preferable in social science for two main reasons: they do not rely on arbitrary thresholds to determine whether two records constitute a match, and they allow post-merge analyses to account for uncertainty introduced during record linkage (Enamorado *et al.* 2019). For applications where there may be multiple correct matches for each observation, a method that can estimate match probabilities will provide a principled approach for determining which records to merge, and how strongly to weight each observation in a subsequent analysis.

In this article, I propose a probabilistic record linkage procedure that incorporates pretrained text embeddings into an active learning algorithm (Bosley *et al.* 2025; Enamorado 2018). The approach, which I call `fuzzylink`, is a variant of Adaptive Fuzzy String Matching (Kaufman and Klevs 2022), an iterative process of fitting a model, labeling uncertain matches, refining the model, and repeating until the model converges. The labeling step is performed by zero-shot prompts to a language model, which reduces time and expense compared to hand-labeling (Ornstein, Blasingame, and Truscott 2025). Across a series of political science applications, I show that this approach significantly improves both precision and recall over existing approaches, and can even perform some tasks—like multilingual record linkage—that would be impossible using lexical similarity measures alone. In this article, I focus on applications with a single fuzzy matching variable (and potentially multiple exact “blocking” variables), and conclude by discussing how one might extend the procedure to multiple fuzzy matching variables.

2. The Algorithm

Consider the problem of merging two datasets $\mathcal{A}$ and $\mathcal{B}$, with sample sizes $n_{\mathcal{A}}$ and $n_{\mathcal{B}}$, respectively. Let X be a matrix of predictors measuring the similarity between each record pair in the set $\mathcal{A} \times \mathcal{B}$. The goal of a probabilistic record linkage procedure is to estimate a model $f(X)$ that maps X onto a match probability for each record pair.

The workhorse model for this class of problem was first formalized by Fellegi and Sunter (1969). The Fellegi–Sunter model is an *unsupervised* approach to record linkage, because it does not require the researcher to provide labeled data on the true matching status of record pairs. Instead, the model estimates match probabilities using unlabeled data, requiring the researcher to pre-specify a set of discrete thresholds for what level of similarity constitutes a match. One advantage of this approach is that it can incorporate similarity metrics from many different types of variables, including strings, numbers, and geographic coordinates. Another key advantage is its computational efficiency. The `fastLink` implementation of the Fellegi–Sunter model by Enamorado *et al.* (2019) can easily handle merging large-scale administrative datasets with hundreds of millions of observations.

However, an unsupervised approach can be an inappropriate choice for record linkage problems with a single fuzzy matching variable—like the applications described in this article—because without overlapping information from multiple matching variables, the model’s accuracy will be quite sensitive to the researcher’s choice of similarity thresholds. In such cases, one will generally prefer a *supervised* approach, in which the model learns the mapping between string similarity and match probability based on labeled data.

The main practical impediment to a supervised approach is that the total number of record pairs scales with $n_{\mathcal{A}} \times n_{\mathcal{B}}$, so it quickly becomes infeasible to label every record pair even in small-scale applications. The general solution to such problems is active learning (Bosley *et al.* 2025; Enamorado 2018). Rather than exhaustively labeling every pair of records, an active learning approach identifies the most *informative* record pairs with which to train the model—that is to say, the record pairs for which the model is most uncertain. By iteratively fitting a model, selecting informative pairs to label, refitting the model, and repeating until the model converges, one can train a supervised learner using a relatively small number of observations.

The active learning algorithm described below performs a fuzzy “left join,” identifying every record in $\mathcal{B}$ that matches at least one record in $\mathcal{A}$. It proceeds in six steps.

Step 1: Embedding. Select the string variable that identifies each record in $\mathcal{A}$ and $\mathcal{B}$, and retrieve text embeddings for each unique string. In the analyses that follow, I use 256-dimensional pretrained embeddings from OpenAI.¹ Wherever possible, the strings representing records should *not* be pre-processed by stemming, converting to lowercase, or any other steps that one might take to reduce

¹The most up-to-date embedding model offered by OpenAI as of February 2025 returns 3,072-dimensional embeddings, but one can reduce the dimensionality through “Matryoshka Representation Learning” (Kusupati *et al.* 2024), dramatically improving computation speed at little cost to accuracy. The most recent training data for these embedding models is September

complexity in a bag-of-words representation (Grimmer and Stewart 2013); performance will generally be improved if we embed text as it is most likely to appear in the training corpus (e.g., “Coca-Cola” instead of “cocacola”). The output from this step will be two matrices $\mathbf{M}_A$ and $\mathbf{M}_B$, with dimensions $n_A \times 256$ and $n_B \times 256$, respectively. Each row of these matrices is an embedding vector.²

Step 2: Compute Similarity Metrics. For each pair of records in the set $\mathcal{A} \times \mathcal{B}$, compute the cosine similarity between their embedding vectors. If the embeddings are normalized to length 1, a matrix of cosine similarities can be efficiently computed by taking the product $\mathbf{M}_A(\mathbf{M}_B)'$. If there are any variables that must match exactly to link a record from $\mathcal{A}$ to $\mathcal{B}$ (“blocking variables”), perform this step only for pairs of records with exact matches on these variables. Since the computational complexity of this step scales with $n_A \times n_B$, blocking can significantly improve efficiency and as practical matter should be used whenever possible.

Step 3: Label a Training Set. Select a subset of record pairs and assign each pair a binary label, 1 if the records are a true match and 0 otherwise. For this article’s analyses, I begin with an initial training set of the 500 record pairs with the highest cosine similarity scores and generate labels using the following zero-shot prompt to OpenAI’s GPT-4o³:

Decide if the following two names refer to the same {record_type}.
{additional_instructions} Think carefully.⁴ Respond with "Yes" or
"No".

Name A: {A}

Name B: {B}

The placeholders {record_type} and {additional_instructions} will vary by application. The accuracy of LLM labels is often improved by including context-specific instructions or examples (Ornstein, Blasingame, and Truscott 2025), just as a researcher would include a detailed codebook if this step were conducted by human research assistants or crowd-coders.

Step 4: Fit Supervised Learner. Fit a probabilistic model to map these cosine similarities onto a match probability. In the analyses that follow, I fit a logistic regression, which has the advantage of being significantly faster at generating predictions for large datasets than other supervised learners. I include as predictors both embedding similarity and Jaro-Winkler similarity, to capture both semantic and lexical differences between records.⁵

Step 5: Label Informative Cases. Estimate match probabilities for all record pairs in the set $\mathcal{A} \times \mathcal{B}$ using the fitted model from Step 4. Select N_L record pairs to label using uncertainty sampling, where selection probability is determined by a Gaussian kernel centered on match probability of $\frac{1}{2}$ (Enamorado 2018).⁶ Assign labels to these record pairs as in Step 3. Add the new labeled observations to the training set and refit the model as in Step 4. Repeat these steps until a stopping criterion is met. In this article’s analyses, I label $N_L = 100$ record pairs per active learning iteration, and stop when none of the estimated probabilities $f(X)$ change by more than 0.01 between iterations.

2021, meaning the approach will underperform if successfully linking records requires knowledge of events that have occurred since that date.

² An alternative approach to computing embedding similarity—called cross-encoders (Lin 2025)—is to pass string pairs directly to a transformer model, outputting a similarity score. Although such an approach could improve accuracy, it would come at the cost of quadratic computational complexity during the embedding step, so I do not implement it here.

³ In the Supplementary Material, I replicate the article’s empirical applications using open-source language models for the embedding and labeling steps. The advantage of open-source models is that their results are fully reproducible, though this comes at the expense of poorer record linkage accuracy. I discuss this tradeoff more fully in Section 4.

⁴ Bizarre as it may seem, prompts that include phrases like “Think carefully” often yield marginal gains in classification accuracy (Battle and Gollapudi 2024).

⁵ In Section A.1 of the Supplementary Material, I vary the model specification in Step 4 and show that this choice yields the best-calibrated probability estimates.

⁶ Formally, selection probability is based on estimated log-odds. I use the kernel $\mathcal{N}(0, 0.2)$, which has 95% of its mass between 40% and 60% match probability.

Step 6: Link Datasets. If the number of labeled record pairs is fewer than a researcher-specified budget $N_{\max}$, continue labeling record pairs from $\mathcal{A}$ without an identified match in $\mathcal{B}$. This will generally improve recall by identifying true matches with low estimated match probabilities. Return all record pairs and their estimated match probability. Optionally, only return record pairs with an estimated match probability greater than π , where π is selected to maximize expected F_1 score.⁷

3. Applications

In this section, I describe four applications of the method, testing its performance across a variety of record linkage tasks common in political science. The first application merges the names of over 9,000 candidates for public office with voter file records from tens of millions of registered voters in California. The second application replicates an analysis merging misspelled names of U.S. cities with a dataset of place names from the U.S. Census Bureau. The third application merges the names of interest groups with ideology scores estimated from campaign contributions. And the final application explores how well the method can perform record linkage across multiple languages, merging the names of political parties from 32 countries in 30 different languages.

For each application, I evaluate performance by computing both precision and recall, where precision measures the fraction of identified matches that are correct, and recall measures the fraction of correct matches that are identified.

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}.$$

Any method that performs well on both of these metrics is likely to be particularly useful for researchers. Higher precision increases the quality of matches, reducing measurement error in subsequent empirical analyses. Higher recall reduces the amount of missing data in the linked dataset, increasing the statistical power of downstream analyses (Kaufman and Klevis 2022)—as well as reducing bias whenever that missingness is non-random.

3.1. Linking Candidates to Voter File Records

Every year, hundreds of thousands of candidates are elected to local public office throughout the United States. Collecting data on these elections can be a painstaking process (de Benedictis-Kessner *et al.* 2023; Einstein, Ornstein, and Palmer 2022; Sumner, Farris, and Holman 2020), because unlike candidates for state and federal office, there is often very little information recorded about local candidates except their names. In this application, I merge the names of every candidate for mayor and city council in the state of California since 2016 with their corresponding records in the L2 voter file. There are a total of 9,025 unique candidate names, and roughly 22 million registered voters in the California voter file.⁸ I merge these two datasets using full name as the fuzzy matching string and exact blocking on last name and city of residence. To make validation feasible, the author and a research assistant hand-coded matches from three counties—Alameda, Kern, and Ventura—to estimate precision and recall.

Of the 840 candidates that ran for office in these three counties, `fuzzylink` identified 770 potential matches in the voter file. 154 of these were exact matches, and the research team determined that 584 of the remaining fuzzy matches were valid, for an estimated precision of 95.8%. In addition, the research

⁷The F_1 score is the harmonic mean of precision and recall, as defined in Section 3. This step will typically remove a large number of record pairs to which the model assigns a very low match probability, significantly reducing the difficulty of post-merge manual validation (Section 4).

⁸The California Election Data Archive (CEDA) is available at <http://www.csus.edu/isr/projects/ceda.html>.

team was able to locate 32 matches in the L2 voter file that *fuzzylink* failed to identify,⁹ for an estimated recall rate of 95.8%. By comparison, the *fastLink* approach (Enamorado *et al.* 2019)—which links records based on predetermined cutoffs in Jaro-Winkler scores¹⁰—identifies only 521 potential matches, with an estimated precision of 93.3% and recall of 63.1%. The dramatically improved recall is largely due to *fuzzylink* successfully linking a variety of nicknames from the candidate list with legal names in the voter file (e.g., “Vinnie” with “Vinton,” “Chuck” with “Charles,” “Libby” with “Elizabeth,” “Trish” with “Patricia,” “Mel” with “Carmelita,” “Sri” with “Sricharana,” and “Teddy” with “Theadora”). There are also a number of cases where candidates go by their middle name (e.g., “Gregory Tod Abbott” listed as “Tod Abbott” on the ballot) and are correctly paired by the LLM prompt.

It is worth noting, in light of ongoing debates over algorithmic bias in language models (Abid, Farooqi, and Zou 2021; Grossmann *et al.* 2023), that a disproportionate share of false positive matches (26 out of 32) are Asian, Hispanic, or African American names. As with any record linkage procedure, researchers should take the time to carefully examine a subset of the merged dataset and ensure that the method is performing as expected. Fortunately, the estimated match probabilities are well-calibrated (see Section A.1 of the Supplementary Material) and can serve as a useful guide during validation: the false positives had a median match probability of just 22%, compared to 58% for the true positives. One could eliminate over half of the false positives in the merged dataset by manually validating only the 188 least-probable matches.

3.2. Linking Misspelled City Names to U.S. Census Bureau Records

Next, I replicate a record linkage task from Kaufman and Klevs (2022), which allows for a direct comparison between the two approaches. There are four key differences between the *fuzzylink* algorithm and the AFSM algorithm proposed by Kaufman and Klevs (2022): (1) the inclusion of embedding similarity as a predictor of match quality, (2) the use of automated labeling by LLMs instead of human coders, (3) a logistic regression classifier instead of random forest, and (4) selecting record pairs to label through uncertainty sampling. See Section A of the Supplementary Material for a detailed ablation analysis, exploring the effect of each of these choices.

The first dataset contains information on 661,218 loan recipients from the 2021 Paycheck Protection Program (PPP) implemented by the U.S. federal government in the wake of the COVID-19 pandemic. This dataset contains each recipient’s address, but the city name provided is rarely an exact match with place names as listed by the U.S. Census Bureau. If a researcher wanted to determine which U.S. municipalities were receiving funds through this program, it would require linking 7,118 misspelled city names in the PPP records with a place-level dataset of 28,889 incorporated towns and cities maintained by the Census, blocking by state.

This is another application where methods that rely on lexical similarity alone can fall short, because many pairs of cities have quite similar names, potentially yielding a large number of false positive matches. Replicating the AFSM procedure as described in Kaufman and Klevs (2022) yields a set of 1,075 city pairs for which the model assigns a match probability greater than 90%. Of these record pairs, only 705 were confirmed as true matches by the research team, for an estimated precision of 66%. Some examples of the 370 false positive matches include Wingdale NY → Walden NY, Deerpark TX → Parker TX, Maple TX → Palmer TX, Lamont MI → Almont MI, Delair NJ → Garfield NJ, and Malone WI → Montreal WI.

Adding embedding similarity as a predictor in the AFSM algorithm significantly improves performance, increasing precision to 84% and the number of matches correctly identified to 1,049 (see Table 2). But despite this improvement, the approach still struggles with more challenging cases, yielding a number of false positive matches like Saint Augustine FL → Saint Augustine Beach FL, Preston CT → New Preston CT, and Swanzey NH → West Swanzey NH.

⁹This search was conducted with the aid of local newspaper articles, campaign websites, and obituaries.

¹⁰I use the package’s default thresholds of 0.88 for a partial match and 0.94 for a full match.

Table 2. Performance metrics for city name merge across three algorithms.

Algorithm	True matches identified	Precision
AFSM	705	65.6%
AFSM with embeddings	1,049	84.1%
fuzzylink	2,451	98.0%

The *fuzzylink* algorithm dramatically improves over both these approaches, more than doubling the number of true matches recalled with near-perfect precision. Key to this performance is the combination of uncertainty sampling and accurate LLM labeling, so that challenging cases like the ones mentioned above are identified and correctly labeled by the LLM during the active learning loop. Not only is precision improved by removing these false positives, but recall is improved by identifying true matches with low lexical similarity. These include OKC → Oklahoma City, Olympic Valley WA → Squaw Valley WA, and USAF Academy CO → Air Force Academy CO. The algorithm also correctly pairs cases where the PPP loan recipient listed a neighborhood rather than a city in their address, like Astoria NY → Queens NY, Newbury Park CA → Thousand Oaks CA, and Port Bolivar TX → Bolivar Peninsula TX.

3.3. Linking Amicus Cosigners to Campaign Donations

For the article's third application, I replicate the record linkage from Abi-Hassan *et al.* (2023), who estimate the ideology of interest groups by merging the names of organizations that cosigned Supreme Court amicus curiae briefs (Box-Steffensmeier, Christenson, and Hitt 2013) with ideal point estimates (DIME scores) from campaign donations (Bonica 2014). There are 15,376 organizations in their dataset and 2.9 million organizations with recorded campaign donations in the DIME dataset. The scale of these datasets poses a significant practical challenge for computation and validation—without any blocking variables, linking the full versions of both datasets requires computing approximately 38 billion pairwise similarity scores. To make manual validation feasible, I focus here on the 1,388 organizations that cosigned amicus briefs in the year 2012, and to reduce computational complexity, I also restrict the DIME dataset to organizations with at least eight distinct campaign contributions. This is both a practical and principled choice, since “donating to eight or more distinct recipients is [typically] sufficient to recover a reliable ideal point estimate” (Bonica 2023).

Through a combination of exact matching and fuzzy string matching, Abi-Hassan *et al.* (2023) were able to locate DIME scores for 376 of these 1,388 organizations, approximately 27% of the total. By comparison, despite restricting its search to only 8% of the DIME dataset, *fuzzylink* is able to locate DIME scores for 437 unique organizations. As in the first application, this dramatically improved recall is largely the result of correctly identifying alternative names for the same organization (e.g., “Utah Association for Justice” and the “Utah Trial Lawyers Association,” “California Forestry Association” and “CA Forestry Assoc PAC,” and “Ojibwe” and “Chippewa” tribes) and even former names of the same organization (e.g., “Airlines For America” formerly “Air Transport Association of America,” “California Construction Trucking Association” formerly “California Dump Truck Owners Association,” “United States Telecom Association” formerly “United States Telephone Assn,” and “PacifiCorp” formerly “Pacific Power & Light”). This improved recall does not appear to come at the expense of precision: the research team identified only three false positives out of 939 proposed matches, for an estimated precision of 99.6%. Note that this estimate considers chapters or subsidiaries of larger organizations to be true matches (e.g., linking “NAIOP” and “NAIOP New Jersey Chapter”), under the assumption that one can use campaign donations of local chapters to make inferences about the parent organization's ideology. If one were unwilling to make such an assumption, those matches could easily be filtered out post-merge, or one could modify the LLM prompt in Step 3 to ignore such matches.

Table 3. Multilingual record linkage application: splitting the ParlGov data into two sets with 4,972 observations each.

Native party names:			
country_name	election_date	party_name	seats
Austria	1919-02-16	Sozialdemokratische Partei Österreichs	72
Austria	1919-02-16	Österreichische Volkspartei	69
Austria	1919-02-16	Deutschnationale	8
Austria	1919-02-16	Deutsche Freiheits und Ordnungspartei	5
.	.	.	.
.	.	.	.
Turkey	2023-05-14	Zafer Partisi	0
English party names:			
country_name	election_date	party_name	left_right
Austria	1919-02-16	Social Democratic Party of Austria	3.7293
Austria	1919-02-16	Austrian People's Party	6.4733
Austria	1919-02-16	German-Nationals	7.4000
Austria	1919-02-16	German Freedom and Order Party	8.8000
.	.	.	.
.	.	.	.
Turkey	2023-05-14	Victory Party	8.8000

Note: The first dataset contains each party's name in the country's native language, and the second dataset contains the English name of each political party.

3.4. Linking Political Party Names Across Multiple Languages

For record linkage problems involving multiple languages, lexical similarity measures tend to be a poor guide to match quality. The strings “LDP” and “Jiyū Minshutō,” for example, share no lexical features at all, but both refer to the same Japanese political party. Pretrained text embeddings, by comparison, can naturally accommodate this sort of problem by representing text from multiple languages in the same embedding space. This makes transformer models particularly adept at machine translation tasks (Vaswani *et al.* 2017). In this application, I demonstrate that the approach proposed here can successfully link the names of political parties across 30 languages—though performance is better for some languages than for others.

To test the method, I take the ParlGov dataset of parliamentary elections since 1900 (Döring and Manow 2018), splitting it into two datasets as illustrated in Table 3. The first dataset contains each party’s name in its native language, the election year, and the number of seats the party won in parliament that year. The second dataset contains the English translation of the party’s name along with its estimated left-right ideology on a scale from 0 (leftmost) to 10 (rightmost). I include all parties from non-English speaking countries that won seats in parliament, for a total of 4,972 observations across 32 countries and 663 elections. Because text embeddings may be closer in space for some language pairs than others,¹¹ I perform this record linkage separately for each country, blocking on election date.

¹¹For example, as measured by cosine similarity, the phrase “Social Democratic Party” is much closer to the Portuguese “Partido Social Democrata” (0.80) than it is to the Icelandic Social Democratic Party “Alýflokkin” (0.44). However, “Alýflokkin” is closest to “Social Democratic Party” relative to other Icelandic parties, so the probabilistic model will perform best if we avoid pooling embedding distances across language pairs.

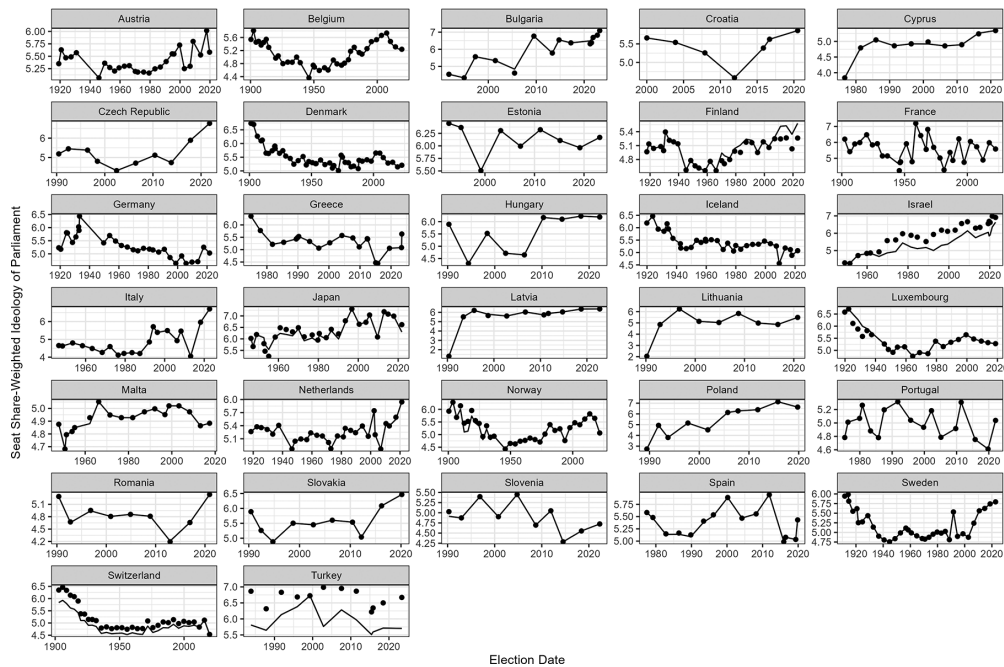


Figure 1. Estimated seat-weighted parliamentary ideology following merge (points) plotted over true values (lines).

The resulting dataset correctly matches 4,855 name pairs out of 4,972—a recall rate of 97.6%. There are, however, a large number of false positive matches (453 in total), for an overall precision of 91.5%. As expected, the method's accuracy varies somewhat by language: precision and recall are lower for countries like Israel (68.9% precision, 87.0% recall) and Japan (79.1% precision and 94.6% recall) than for Italy (98.3% precision, 99.6% recall) or Portugal (100% precision, 100% recall). See Table A4 in the Supplementary Material for a complete list of these evaluation metrics by country.

In addition to computing these accuracy metrics, one can evaluate whether the record linkage procedure allows us to recover downstream quantities of interest. Figure 1 plots the seat-share weighted ideology of every parliament in the ParlGov dataset (lines) along with each parliament's estimated ideology following the record linkage (points).¹² The correlation between the estimates and their true values is 0.964, and the estimates are perfectly correlated with the truth in most countries. Only a few country-years stand out as severely mis-estimated. In Switzerland, the model incorrectly links the FDP ("Freisinnig-Demokratische Partei der Schweiz") with both the Liberal Party of Switzerland and the Radical Democratic Party. These two parties merged in 2009, but they were separate parties throughout the prior century, which biases our estimates rightwards for much of the 20th century. In Israel, the model incorrectly links the Labor Party ("HaAvoda") with the right-wing Likud, and in Turkey, the model fails to identify a match for the Social Democratic Populist Party. In both cases, these errors bias the estimates rightward.

In practice, errors like these can be easily corrected by conducting a post-merge manual validation, focusing on records in $\mathcal{A}$ that did not match to a single unique record in $\mathcal{B}$. In this case, it would require manually checking only 128 proposed matches, roughly 2% of the total.

¹²For each party in $\mathcal{A}$, estimated ideology is computed as the average ideology of its matches in $\mathcal{B}$, weighted by match probability.

4. Discussion

The approach I propose here has significant advantages over methods that rely on lexical similarity measures alone. Social scientists often encounter record linkage problems where matching entities may be lexically dissimilar from one another, whether it's due to alternative names, acronyms, or even different languages. Under such conditions, the `fuzzylink` procedure can significantly improve both precision and recall. And it does so without requiring significant expenditure in time or money. None of the applications described in the previous section took longer than a few hours to execute on a personal computer or cost more than \$10 in API fees (see Section B of the Supplementary Material for a more thorough cost breakdown).

Despite these advantages, there are several situations in which the proposed approach is likely to fall short or prove unnecessary. For example, when merging large-scale administrative datasets with tens or hundreds of millions of records, researchers are likely to prefer the added efficiency of an unsupervised approach like `fastLink` (Enamorado *et al.* 2019). Particularly when such datasets contain a large number of identifying fields, the marginal gains in accuracy from a supervised approach like `fuzzylink` are unlikely to be worth the loss of efficiency. The proposed approach is also likely to fail when pretrained language models do not encode the relevant world knowledge necessary to link two records. The embeddings used in this article, for example, are trained only on data collected before September 2021, and will therefore struggle to perform any record linkage task that requires knowledge of events after that date. Post-merge clerical review is essential to ensure the accuracy of the LLM labels, replacing them with human labels in cases where they perform poorly. Finally, researchers will find this approach unnecessary in applications where discrepancies between records are due solely to typos or misspellings, in which case embedding similarity offers little predictive advantage over lexical similarity alone.

I have focused in this article on applications where there is a single fuzzy string matching variable, but the sorts of record linkage problems faced by social scientists often include many such variables. Fortunately, the method can be extended in a number of ways. One approach would be to re-express multiple fuzzy variables as a single string, which can then be represented as an embedding. For example, a record with {name} and {address} fields might be represented by the string “My name is {name} and I live at {address}.” Another approach would be to estimate a match probability separately for each variable as I have done here, and then use those match probabilities as inputs in the Fellegi–Sunter model (Enamorado *et al.* 2019). Further research is needed to determine which approach yields better results.

Another limitation of the method as presented is its reliance on proprietary language models. Because these models are closed-source and operated by for-profit entities, they can be deprecated or modified at any time without the consent of their users. Consequently, the results that `fuzzylink` produces—including those presented in this article—are not fully reproducible. Though a researcher could replicate the steps I used to generate the results, within a few years, it will be impossible to reproduce them exactly. For this reason, many scholars in our discipline have urged using open-source language models wherever possible (Spirling 2023).

Unfortunately, as of writing, it is difficult to see how the method presented here could be undertaken using open-source language models. Frankly, the level of accuracy I demonstrate here would not have been possible even using the previous generation of *proprietary* language models. In the Supplementary Material, I attempt to replicate the article's empirical applications using one of the highest performing open-source language models currently available (Mistral 8x22B), as well as the previous generation of language models released by OpenAI as of early 2023 (GPT-3.5). These variants significantly underperform the results reported in the previous section, particularly for the organization matching and multilingual record linkage applications. Given the rapid development of open-source language models, it is likely that there will be an acceptable open-source solution in the coming years, but until that time, the accuracy gains from proprietary models outweigh their drawbacks.

When a research method falls short of full computational reproducibility, one must insist that it meet standards of *replicability* (procedures are transparently documented so that other scholars can independently replicate them) and *reliability* (repeated application of the procedure yields similar, if

not identical, outcomes). Indeed, these are the standards that our discipline applies to other non-reproducible research methods, like those that rely on human research assistants or crowd-coders. The `fuzzylink` software package¹³ was developed to help researchers implement the method proposed here in a straightforward and replicable manner, and I hope that it will enable much useful social science research in the coming years.

Acknowledgements. Thanks to Elise Blasingame, Jason Byers, Rob Carroll, David Cottrell, Ted Enamorado, Kosuke Imai, Aaron Kaufman, George Krause, Avital Livny, Isaac Mehlhaff, Brandon Stewart, Maria Tislenko, Garrett Vande Kamp, and participants at Polmeth 2024 and SPSA 2025 for their helpful comments and suggestions.

Data Availability Statement. Replication code for this article is available at Ornstein *et al.* (2025). A preservation copy of the same code and data can also be accessed via Dataverse at <https://doi.org/10.7910/DVN/7U5KEJ>. The `fuzzylink` R package is available for download through the Comprehensive R Archive Network (CRAN).

Competing Interests. The author declares no competing interest.

Supplementary Material. The supplementary material for this article can be found at <https://doi.org/10.1017/pan.2025.10016>.

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¹³Implemented in the R programming language, and available for download through the Comprehensive R Archive Network (CRAN).

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